

Q3 -10

Q1 -20

Q2. -8 A1 equals to 2 also

Q4 Good



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Homework 6 - Deadline December 13th

Exercise 1. 20 points. Find all the differentiable functions $f : (0, +\infty) \rightarrow (0, +\infty)$ such that $f(e/3) = 1$ and

$$\left[x + x \ln^2(3x) \right] f^2(x) f'(x) = f^3(x)$$

Assume $g(x) = f^3(x)$, $g'(x) = 3f^2(x)f'(x)$

1.1

we can get $\left[x + x \ln^2(3x) \right] \frac{1}{3} g'(x) = g(x)$

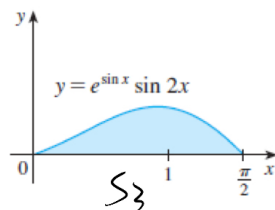
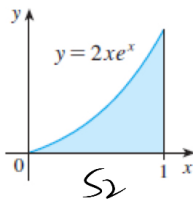
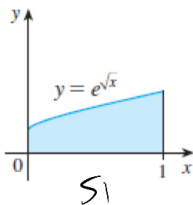
Since $g(\frac{e}{3}) = f^3(\frac{e}{3}) = 1$

$$\left[\frac{e}{3} + \frac{e}{3} \ln^2\left(3 \cdot \frac{e}{3}\right) \right] \frac{1}{3} g'\left(\frac{e}{3}\right) = g\left(\frac{e}{3}\right)$$

$$\frac{2}{3}e \cdot \frac{1}{3} g'\left(\frac{e}{3}\right) = 1$$

$$g'\left(\frac{e}{3}\right) = \frac{9}{2e}$$

Exercise 2. 20 points. Which of these areas are equal? Why? Make all the computations that you need to justify your answer.



$$S_2 = S_3$$

$$(1) y = e^{\sqrt{x}}$$

$$S_1 = \int_0^1 e^{\sqrt{x}} dx$$

$$\text{Assume } u = \sqrt{x}$$

$$du = \frac{1}{2\sqrt{x}} dx$$

$$2\sqrt{x} du = dx$$

$$2u du = dx$$

$$\text{When } x=0, u=0$$

$$x=1, u=1$$

$$\int_0^1 e^{\sqrt{x}} dx = \int_0^1 e^u \cdot 2u du$$

$$= 2 \int_0^1 e^u \cdot u du$$

$$\text{Assume } v = u, du = e^u du$$

$$\text{then } dv = du, u = e^u$$

$$\int e^u \cdot u du = u \cdot e^u - \int e^u du$$

$$= u \cdot e^u - e^u + C$$

$$\text{So } \int e^u \cdot u du = \sqrt{x} \cdot e^{\sqrt{x}} - e^{\sqrt{x}} + C$$

$$S_1 = 2 \int_0^1 e^u \cdot u du = 2[(1 \cdot e - e) - (0 - e)] = 2e$$

$$(2) y = 2xe^x$$

$$S_2 = \int_0^1 2xe^x dx$$

$$\text{Assume } u = 2x, dv = e^x dx$$

$$du = 2dx, v = e^x$$

$$\int 2xe^x dx = 2x \cdot e^x - \int 2e^x dx$$

$$= 2x \cdot e^x - 2e^x + C$$

$$S_2 = \int_0^1 2xe^x dx$$

$$= (2 \cdot e - 2e) - (0 - 2 \cdot 1)$$

$$= 2$$

$$(b) y = e^{\sin x} \sin 2x$$

$$S_3 = \int_0^{\pi/2} e^{\sin x} \sin 2x dx$$

$$= \int_0^{\pi/2} e^{\sin x} \cdot 2 \sin x \cos x dx$$

$$= 2 \int_0^{\pi/2} e^{\sin x} \cdot \sin x \cos x dx$$

$$\text{Assume } u = \sin x$$

$$du = \cos x dx$$

$$\int e^{\sin x} \cdot \sin x \cos x dx$$

$$= \int e^u \cdot u du$$

$$S_3 = 2 \int_0^{\pi/2} e^u \cdot u du$$

$$\text{Assume } v = u, du = e^u du$$

$$dv = du, u = e^u$$

$$\int e^u \cdot u du = u \cdot e^u - \int e^u du$$

$$= u \cdot e^u - e^u + C$$

$$\text{So } \int e^u \cdot u du = \sin x \cdot e^{\sin x} - e^{\sin x} + C$$

$$S_3 = 2 \int_0^{\pi/2} e^u \cdot u du$$

$$= 2[(1 \cdot e^1 - e^1) - (0 - 1)]$$

$$= 2$$

In conclusion, $S_2 = S_3 = 2$

Exercise 3. a) 15 points. Find the area of the region enclosed by the curves

$$f(x) = \frac{x}{10} \text{ and } g(x) = \frac{x}{\sqrt{x^2 + 36}}$$

b) 20 points. Find the volume of the solid obtained by the rotation of the region of part a) around the x -axis.

a) First, we should find the intersections of these two functions, that is:

$$\frac{x}{10} = \frac{x}{\sqrt{x^2 + 36}}$$

$$\text{When } x=0, x^2+36 > 0, \frac{0}{10} = \frac{0}{0}$$

$$\text{then } x \neq 0, \text{ we write } \frac{1}{10} = \frac{1}{\sqrt{x^2 + 36}}$$

$$x^2 + 36 = 100$$

$$x^2 = 64$$

$$x = \pm 8$$

So there are three intersections $x = -8, 0, 8$

The area of the region A is:

$$A = \int_{-8}^0 |g(x) - f(x)| dx + \int_0^8 |g(x) - f(x)| dx$$

$$\text{Since } x \in [-8, 8], \sqrt{x^2 + 36} \leq 10$$

$$A = \int_{-8}^0 \frac{x}{10} - \frac{x}{\sqrt{x^2 + 36}} dx + \int_0^8 \frac{x}{\sqrt{x^2 + 36}} - \frac{x}{10} dx$$

$$= \int_{-8}^0 \frac{x(\sqrt{x^2 + 36} - 10)}{10\sqrt{x^2 + 36}} dx + \int_0^8 \frac{x(10 - \sqrt{x^2 + 36})}{10\sqrt{x^2 + 36}} dx$$

$$\text{Assume } u = x^2 + 36, du = 2x dx$$

$$\text{when } x=0, u=36$$

$$x=\pm 8, u=100$$

$$A = \frac{1}{2} \left(\int_{100}^{36} \frac{\sqrt{u} - 10}{10\sqrt{u}} du + \int_{36}^{100} \frac{10 - \sqrt{u}}{10\sqrt{u}} du \right)$$

$$= \int_{36}^{100} \frac{10 - \sqrt{u}}{10\sqrt{u}} du$$

$$\text{Assume } v = \sqrt{u}, dv = \frac{1}{2\sqrt{u}} du$$

$$\text{when } u=36, v=6$$

$$u=100, v=10$$

$$A = \int_6^{10} (10 - v) \frac{1}{5} dv$$

$$= \frac{1}{5} \int_6^{10} 10 - v dv = \frac{1}{5} \left[10 \times 10 - \frac{1}{2} 10^2 - \left(10 \times 6 - \frac{1}{2} 6^2 \right) \right] = \frac{8}{5}$$

In conclusion, the area is $\frac{8}{5}$

(b) Since $f(x)$ and $g(x)$ are even

$$V = 2 \cdot \pi \int_0^8 [g(x)^2 - f(x)^2] dx$$

$$= 2\pi \int_0^8 \frac{x^2}{x^2 + 36} - \frac{x^2}{100} dx$$

$$= 2\pi \int_0^8 \frac{x^2}{x^2 + 36} dx - 2\pi \int_0^8 \frac{x^2}{100} dx$$

Exercise 4. a) 25 points. Let $f : [0, 1] \rightarrow \mathbb{R}$ be a continuous function. Prove the identity:

$$\int_0^\pi x f(\sin(x)) dx = \frac{\pi}{2} \int_0^\pi f(\sin(x)) dx$$

b) Use part a) to compute

$$\int_0^\pi \frac{x \sin(x)}{1 + \cos^2(x)} dx$$

a) Prove

Assume $u = \pi - x$, $du = -dx$

when $x=0$, $u=\pi$

$x=\pi$, $u=0$

$$\begin{aligned} \int_0^\pi x f(\sin(x)) dx &= \int_\pi^0 (\pi - u) f(\sin(\pi - u)) (-du) \\ &= \int_0^\pi (\pi - u) f(\sin(u)) du \end{aligned}$$

$$\begin{aligned} \text{Then we write } I &= \int_0^\pi x f(\sin(x)) dx + \int_0^\pi (\pi - x) f(\sin(x)) dx \\ &= \int_0^\pi \pi f(\sin(x)) dx \\ &= \pi \int_0^\pi f(\sin(x)) dx \end{aligned}$$

In conclusion, $\int_0^\pi x f(\sin(x)) dx = \frac{\pi}{2} \int_0^\pi f(\sin(x)) dx$

(b) $\int_0^\pi \frac{x \sin(x)}{1 + \cos^2(x)} dx$

Assume $g(x) = \frac{\sin(x)}{1 + \cos^2(x)}$

According to part (a), $\int_0^\pi x g(x) dx = \frac{\pi}{2} \int_0^\pi g(x) dx$

$$= \frac{\pi}{2} \int_0^\pi \frac{\sin(x)}{1 + \cos^2(x)} dx$$

Assume $u = \cos(x)$, $du = -\sin(x) dx$

when $x=0$, $u=1$

$x=\pi$, $u=-1$

$$\begin{aligned} \frac{\pi}{2} \int_0^\pi \frac{\sin(x)}{1 + \cos^2(x)} dx &= \frac{\pi}{2} \int_1^{-1} \frac{1}{1 + u^2} (-du) \\ &= \frac{\pi}{2} \int_{-1}^1 \frac{1}{1 + u^2} du \\ &= \frac{\pi}{2} (\arctan(1) - \arctan(-1)) \\ &= \frac{\pi}{2} \left(\frac{\pi}{4} - \left(-\frac{\pi}{4} \right) \right) \\ &= \frac{\pi^2}{4} \end{aligned}$$

In conclusion, $\int_0^\pi \frac{x \sin(x)}{1 + \cos^2(x)} dx = \frac{\pi^2}{4}$

Index of comments

- 1.1 rewrite the equation as $f'(x)/(f(x))=1/[x+x((\ln(3x))^2)]$,
 $\ln|f(x)|=\arctan(\ln(3x))+C$
 use initial condition to get C,
 $f(x)=e^{(\arctan(\ln(3x))-\pi/4)}$